

NLP Course Project

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MULTITASK BERT FOR STUDENT SUPPORT UNDERSTANDING

A self-implemented BERT encoder for SST, paraphrase detection, and STS

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1. Problem Statement
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1. Problem Statement

Student-support systems often receive short natural-language requests that require different types of understanding.

Core question

Can one shared BERT encoder support sentiment classification, duplicate-question detection, and semantic similarity at the same time?

- ▶ **Sentiment:** detect the tone of a student message.
- ▶ **Paraphrase:** detect duplicate or equivalent questions.
- ▶ **Similarity:** retrieve semantically related handbook entries.

2. Tasks and Datasets

Task	Dataset	Metric
Sentiment classification	SST-5	Accuracy
Paraphrase detection	Quora Question Pairs	Accuracy
Semantic similarity	STS-B	Pearson correlation

The three tasks are different in output format, but all depend on contextual sentence representations.

3. Proposed Approach

Shared representation

- ▶ One BERT encoder.
- ▶ Three task-specific heads.
- ▶ Joint multitask fine-tuning.

The HuggingFace backend is used only as a sanity-check reference.

Main contribution

We re-implemented the BERT encoder ourselves instead of using `AutoModel` in the main forward path.

4. Why Multitask Learning?

- ▶ SST learns single-sentence sentiment.
- ▶ Quora learns binary semantic equivalence.
- ▶ STS learns graded semantic similarity.
- ▶ Para and STS share sentence-pair structure.

Hypothesis

Sharing the encoder and relational sentence-pair features can improve representation learning across related tasks.

5. BERT Encoder Architecture

Core blocks

- ▶ Token ids
- ▶ Token embeddings
- ▶ Position embeddings
- ▶ Segment embeddings
- ▶ Self-attention
- ▶ Feed-forward
- ▶ Residual + LayerNorm

Diagram placeholder

- ▶ Insert BERT diagram.
- ▶ Input → embeddings.
- ▶ Encoder stack.
- ▶ [CLS] output.

6. Self-Implementation Validation

Parity check against HuggingFace

Same input and same pretrained weights:

- ▶ Max hidden-state difference: 4.29×10^{-6} .
- ▶ Max pooler-output difference: 9.54×10^{-7} .

This validates that the custom encoder is numerically faithful before fine-tuning.

7. Multitask Architecture

Component	Role	Output
Shared BERT encoder	Contextual representation	[CLS] vector
SST head	Sentiment classifier	5 logits
Paraphrase head	Duplicate detector	1 logit
STS head	Similarity regressor	score in $[0, 5]$
Relational layer	Shared Para–STS features	pair interaction

The architecture separates task-specific decisions while keeping the language representation shared.

8. Rich Relational Layer

Para and STS are both sentence-pair tasks. We therefore share an interaction representation before branching into task-specific heads.

$$z = \frac{h_u \odot h_v}{\|h_u\|_2 \|h_v\|_2 + \epsilon}$$

- ▶ h_u, h_v : pooled sentence vectors.
- ▶ Element-wise interaction keeps dimension-level information.
- ▶ STS can benefit from the larger Quora signal.

9. Training Pipeline

Pipeline

- ▶ Load tokenizer
- ▶ Load BERT weights
- ▶ Build dataloaders
- ▶ Interleaved batches
- ▶ Task losses
- ▶ Save best checkpoint

Diagram placeholder

- ▶ Insert training flow.
- ▶ SST / Para / STS.
- ▶ Shared encoder.
- ▶ Task heads.
- ▶ Aggregate score.

Best recipe

- ▶ Optimizer: AdamW.
- ▶ Learning rate: $2e^{-5}$.
- ▶ Weight decay: 0.01.
- ▶ Epochs: 20.
- ▶ Schedule: linear warmup/decay.

Checkpoints are selected by aggregate development score.

11. Evaluation Protocol

Each task is evaluated with its standard metric:

- ▶ SST: accuracy.
- ▶ Paraphrase: accuracy.
- ▶ STS: Pearson correlation.

Aggregate score:

$$S = \frac{SST_{acc} + Para_{acc} + (STS_r + 1)/2}{3}$$

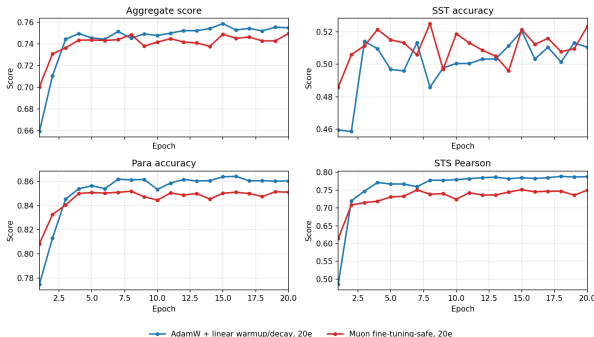
12. Main Results

Model	SST	Para	STS	Score
Self-impl AdamW linear 20e	0.520	0.864	0.785	0.759
HF sanity baseline 10e	0.526	0.861	0.790	0.761
Self-impl AdamW constant LR	0.505	0.860	0.784	0.752
Self-impl Muon safe 20e	0.523	0.851	0.750	0.750
Muon high-LR early run	0.262	0.720	0.145	0.518

Our self-implemented encoder reaches 0.759, nearly matching the library-backed baseline at 0.761.

13. Optimizer Finding

AdamW vs Muon Training History (20 Epochs)



Conclusion

- ▶ AdamW is clearly better.
- ▶ Muon damages pretrained weights.
- ▶ Para / STS drop sharply.
- ▶ Final choice: AdamW.

14. Error Analysis

Current weakness

STS is the weakest task: Pearson $r = 0.785$.

To complete

Insert 2–3 real STS error examples:

- ▶ sentence A / sentence B,
- ▶ gold score / predicted score,
- ▶ short error explanation.

15. Demo A: Checkpoint Inference

Purpose

Directly load one checkpoint and test:

- ▶ SST sentiment.
- ▶ Paraphrase probability.
- ▶ STS similarity score.

`scripts/demo_gradio.py`

Screenshot Placeholder

Replace with checkpoint inference
screenshot.

16. Demo B: Smart Student Support

Product workflow

- ▶ Detect student sentiment.
- ▶ Match duplicate FAQs.
- ▶ Rank handbook entries.

`scripts/demo_student_support.py`

Screenshot Placeholder

Replace with Smart Student Support
screenshot.

17. Demo Commands

Checkpoint inference

```
PYTHONPATH=src python -m scripts.demo_gradio \  
-config configs/report_best_self_impl_20e_linear_warmup_lr2e5.yaml \  
-checkpoint runs/report_best_self_impl_20e_linear_warmup_lr2e5/best.pt
```

Smart student support

```
PYTHONPATH=src python -m scripts.demo_student_support \  
-config configs/report_best_self_impl_20e_linear_warmup_lr2e5.yaml \  
-checkpoint runs/report_best_self_impl_20e_linear_warmup_lr2e5/best.pt
```

18. Conclusions

- ▶ We re-implemented the BERT encoder.
- ▶ Encoder parity is extremely close: 4.29×10^{-6} .
- ▶ Self-implemented score: **0.759**.
- ▶ HF baseline score: 0.761.
- ▶ AdamW clearly outperforms Muon.
- ▶ Two demos connect benchmarks to usage.

Future work: stronger STS calibration, dynamic task weighting, and richer sentence-pair modeling.

THANK YOU

Project: Multitask BERT for Student Support Understanding

Repository: multitask-bert-v2

Q&A

The logo for HUST (Ho Chi Minh University of Science) is displayed in white, bold, sans-serif capital letters. It is centered within a dark blue rectangular area. Surrounding the text is a decorative pattern of red dots of varying sizes, arranged in a circular, spiral-like formation that creates a sense of depth and movement.

- [1] Devlin et al., *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*, 2019.
- [2] Liu et al., *Multi-Task Deep Neural Networks for Natural Language Understanding*, 2019.
- [3] Reimers and Gurevych, *Sentence-BERT*, 2019.
- [4] Loshchilov and Hutter, *Decoupled Weight Decay Regularization*, 2019.